UNIT-4: Geometric Algorithms & Randomized Algorithms

Dt: 07/10/25

Line Segment Properties: 5M/7M/8M

↳ 2M

↳ Geometry Algo

1) Cross Product : 2M

for Vectors $P_1 = (x_1, y_1)$

$P_2 = (x_2, y_2)$

$$P_1 \times P_2 = x_1 \cdot y_2 - x_2 \cdot y_1$$

Example: $P_1 = (1,1)$, $P_2 = (3,1)$

$$P_1 \times P_2 = (1 \times 1) - (3 \times 2) = -5$$

Positive: P_1 is clockwise from P_2

Negative: P_1 is Counterclockwise from P_2

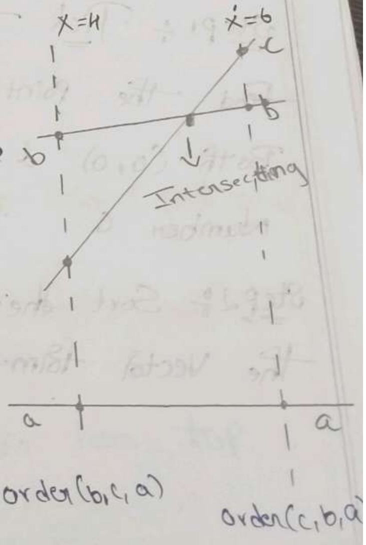
Zero: P_1 & P_2 are Co-linear. → same plane./sameLine.

2) Sweeping Line Technique: 2M

In a given pair does any pair intersect &

- If two segment intersect, then there is some position of the sweep line, just before the intersection point, where those two segments are Consecutive (→ Neighbors / Nearer to each other.) in the Sweep-line Ordering.

(ఇంటర్సెక్ట్ అయిన తర్వాత points దగ్గర ఆర్డర్ ఛేంజ్ అవుతుంది).



Convex Hull

It Computes Smallest Convex polygon that encloses of points in a 2D Plane

Graham Scan Algorithm

The Graham Scan is an efficient method for Computing the Convex hull of a finite Set of 2D points.

Example :-

Solve the given example Using Graham Scan Algorithm also plot the Convex hull.

Points (0,0),(4,0), (4,4)

(0,4), (2,1)

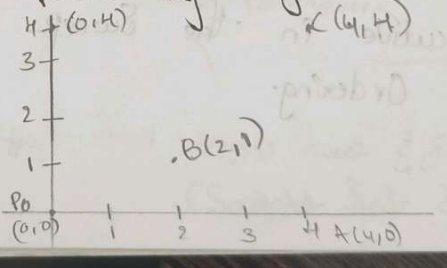
Soln :-

Step1 :- Pick the Starting Point

Find the Point with the Smallest Value

Both (0,0) & (4,0) & (0,4) have Small-Number 0 take $P_0 = (0,0)$

Step 2 :- Sort the point by angle & Calculate the Vector form.



Point	Vector Form	Angle
A(4,0)	(4−0) (0−0) = 4, 0	straight 0°
B(2,1)	(2−0) (1−0) = (2,1)	A little up
C(4,4)	(4−0) (4−0) = (4,4)	Diagonal 45°
D(0,4)	(0−0), (4−0) = (0,4)	straight 90°

Step 3 : Put first 3 elements into the stack
P0, A, B

But P_0 will always first element hence next three points will be
A, B & C

Vector : $C - A = (4-4)\ (4-0) = (0,4)$

$B - A = (2-4)\ (1-0) = (-2,1)$

Cross product $= ((C-A) \times (B-A))$

$= (0,4) \times (-2 \times 1)$

$= (0 \times 1) - (-2)(4)$

$= 8$

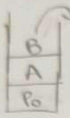
Negative → left → keep the point & push new Point.

Positive → right → Pop top & check previous pts Availability.

Zero → straight line → Pop from top

From Step 3 – Answer is positive hence pop – top which is B & check availability of previous points $P_0 \to A \to C$

Dt : 9/10/25



Step 4 :

$P_0, A, B \longrightarrow C \longrightarrow D$

Now check P_0, A, C

Vector $C - P_0 = (4-0) - (4-0)$
$= (4, 4)$

vector $A - P_0 = (4-0) - (0-0)$
$= (4, 0)$

Cross product $= 4 \times 0 - 4 \times 4$
$= 0 - 16$
$= -16$

Negative → left turn → keep A Push C on the stack.

∴ Current Stack is $P_0 \longrightarrow A \longrightarrow C$

Step 5 : Process $D(0, 4)$

Now $A \longrightarrow C \longrightarrow D$

Vector $D - A = (0-4), (4-0)$
$= (-4, 4)$

Vector $C - A = (4-4), (4-0)$
$= (0, 4)$

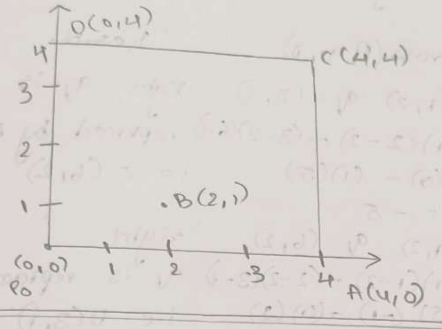
D cross product $= (-4 \times 4) - (0 \times 4)$
$= -16 - 0$
$= -16$

Negative → left → push D & keep C

Current Stack P_0, A, C, D

Step 6 :
~~the~~ h
Conclude
A (4, 0)
inside

Step 6 :- Now all Points are Completed & stack holds the hull. Hence from the given stack it is Concluded that Points hold the hull is $P_0(0,0)$, $A(4,0)$, $C(4,4)$, $D(0,4)$ & Point $B(2,1)$ is inside the Convex hull



Jarvis March :-

It start at the leftmost Point, then keep Picking the next Point so that every other point lies to the left of the line you just drew.

(*) Orientation test for p, q, r

$$\boxed{\text{Cross} = (q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x)}$$

Cross > 0 r is to left $p \rightarrow q$ (counterclock)

Cross < 0 r is to right (r replace q)

↳ q ని తీసి r పెట్టాలి

Cross = 0 Collinear.

Example :- Solve the give example using Jarvis March.

A(1, 2), B(3, 3), C(6, 2), D(3, 1), E(5, 5)

F(4, 2), G(2, 5)

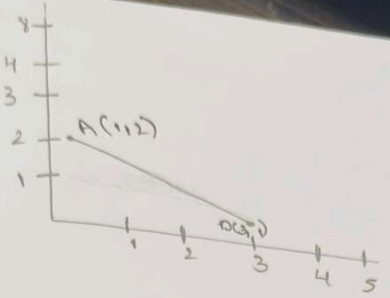
Solution:-

Step 1: Take leftmost point A (1, 2).
Take the very first guess q = B (3, 3) &
for r test every other point.

r	cross (p, q, r)	Result.
C (6, 2)	p = (1, 2) q = (3, 3) (3−1)(2−2) − (3−2)(6−1) (2)(0) − (1)(5) = −5	right q is replaced by r i.e C (6, 2)
D (3, 1)	p = (1, 2) q = (6, 2) (6−1)(1−2) − (2−2)(3−1) (5)(−1) − (0)(2) = −5	right q is replaced by r i.e D (3, 1)
E (5, 5)	p = (1, 2), q = (3, 1) (3−1)(5−2) − (1−2)(5−1) = (2)(3) − (−1)(4) = 6 − (−4) = 10	left keep D
F (4, 2)	p (1, 2) q (3, 1) = (3−1)(2−2) − (1−2)(4−1) = (−2)(0) − (−1)(3) = 0 − (−3) = 3	Keep D
G (2, 5)	p (1, 2), q (3, 1) (3−1)(5−2) − (1−2)(2−1) = (2)(3) − (−1)(1) = 6 − (−1) = 7	Left keep D

Iteration 1
  x y
p (1, 2) A
q (3, 3) B
r (6, 2)

ఇక్కడ left most ~~[illegible]~~ పాయింట్ A ని P గా తీసుకుంటే దాని పక్కనిది q తీసుకుంటాం. q పక్కనిది r ఉన్నాయి. r కోసం మొత్తం కనుక్కోవాలి.


8
4
3
2
1
A(1,2)
D(3,1)
1
2
3
4
5